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LLM-Generated Text in Swedish: Linguistic Features and Detection Methods

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Abstract

This thesis investigates investigate systematic differences between text generated by large language models (LLMs) and human-written text in the Swedish language. In addition, the thesis analyses significant differences in the ability to detect LLM-generated Swedish text between humans and AI-detection tools. The study will include both a formal and informal text component, that is, linguistic analyses and AI-detection experiments will be performed on two sources of data from formal and informal domains.

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Preface

I would like to thank the following people...

1 Introduction

The intended purpose for this study is to investigate systematic differences (if any) between text generated by large language models (LLMs) and text written by humans in the Swedish language. In addition, the study will investigate whether there are any significant differences in the ability to detect LLM-generated Swedish text between humans and LLM tools. The study will include both a formal and informal text component, that is, linguistic analyses and AI-detection experiments will be performed on two sources of data from formal and informal domains.

1.1 Research questions

1. Are there systematic differences between LLM-generated and human-written formal text in Swedish?
2. Are there systematic differences between LLM-generated and human-written informal text in Swedish?
3. Are there differences between humans and AI-based detection tools in their ability to detect LLM-generated Swedish text?

2 Background

This chapter will provide an overview of previous studies related to 1) LLM text generation abilities, 2) human-written and AI-generated comparison studies, and 3) the LLMs of focus in this thesis.

2.1 Large Language Models and text generation

2.1.1 Transformer architectural development

The underlying architecture of LLMs, that is, the Transformer architecture, was first described in Vaswani et al. (2017). In this study, the authors connected an encoder and decoder component of a sequence transduction model to create a simpler network architecture that demonstrates superior performance and less training time on a multimodal range of tasks, notably the task of text generation. Since then, several LLMs have been built upon the Transformer architectural model

Specifically look at LLM architecture and text generation itself, its relationship with training on data.

2.1.2 LLMs for the English language

2.1.3 LLMs for the Swedish language

Holmstrom et al. (2023) initiated an investigation of the extent to which pre-training GPT-3 LLMs on Swedish only is advantageous over leveraging pre-trained knowledge (from English). It also examined whether English-based pre-trained models demonstrated unwanted transfer of Anglophone cultural aspects over to the Swedish context. The findings of Holmstrom et al. (2023) demonstrated that LLMs exposed to limited Swedish during training can be highly capable and transfer competencies from English "off-the-shelf", including mathematical reasoning and Swedish cultural adaptation. In addition, the study showed that GPT-3 models did not appear to exhibit unwanted over-generalisation of cultural aspects to Swedish.

2.2 Linguistic characteristics of LLM-generated text

Several studies have already done that analyse the linguistic characteristics of LLM-generated text in English.

2.2.1 methods of textual analysis

Methods of textual analysis that have existed for decades (?) are now being applied to compare the differences between human-written and AI-generated text. Recent research has focused on English-language texts.

Munoz-Ortiz et al. (2024) compared the linguistic components of human and LLM-generated news articles and compared their patterns among six generative LLMs. For their prompting strategy, they asked the models to generate a news paragraph based on the news article headline its first words. For their linguistic analysis, they compared the human-written and LLM-generated text groups through differences in their linguistic structures, including vocabulary distribution, part-of-speech tag distributions, dependency arc lengths, constituent lengths, cosine similarity, and emotional category tagging. The results of this analysis showed significant differences between the human-written and LLM-generated text on all textual analysis metrics. At the lexical level, 5 of the 6 LLM models outputted text with a more restricted vocabulary than the human-written text. For semantics, the models were found to have less of an inclination to display aggressive negative emotions (such as fear or anger) than humans.

Laio et al. (2023) analysed the differences between human-written and ChatGPT-generated text within the medical domain. Medical texts were compared by their vocabulary distribution, part-of-speech tag distribution, dependency arcs, sentiment analysis, and perplexity. In addition, they implemented machine learning methods to detect ChatGPT-generated medical text. The results of the study found that human-written medical texts contained more concrete information, had higher vocabulary diversity, and contained information that human experts would consider more useful. The ChatGPT-generated medical texts appeared more fluent and logical, but expressed more generalised terminology rather than specific, effective information for the problem's context. Overall, this would point to the human-written medical text being more tailored to the situational modelling of the medical case (situational modelling defined in Mahowald et al., 2024) rather than the LLM-generated text.

A Munoz-Ortiz et al. Contrasting Linguistic Patterns in Human and LLM-Generated News Text (2024)

Text domain: News articles

Linguistic analysis: Human texts exhibit more scattered sentence length distributions, more variety of vocabulary, distinct use of dependency and constituent types, shorter constituents, more optimised dependency distances

Sentiment analysis: Humans tend to exhibit stronger negative emotions (fear and disgust), less joy compared to LLM text

Six generative LLMs: Mistral, Falcon, LLaMa models

Human-generated news, ask the models to generate a news paragraph based on the headline and first words of the news.

Linguistic analysis: Differences in distribution of vocabulary, sentence length, POS tags, syntactic structures, psychometric features (tone and emotions)

Linguistic tools - Stanza for segmentation, tokenization, POS tagging, dependency and constituent parsing (good for English)

Analyse if disparities between human and AI change across models of different sizes and families

2.2.2 by genre

2.3 LLM-detection in generated text

2.3.1 LLM-detection for English text

Previous work has also shown evidence of a distinction between the reasoning patterns of LLM tools when compared to those of humans. (CITE) evaluated the medical

reasoning capabilities of DeepSeek R1, computing its outputs to the reasoning patterns of medical domain human experts. The study performed qualitative and quantitative analyses of AI- and human-medical reasoning for 100 clinical cases from the MedQA dataset. The study found a 93% diagnostic accuracy for DeepSeek R1's performance. Many of the LLM's errors were attributed to over-focus on the details of a medical problem, rather than a more holistic and intuitive (?) understanding by humans, as well as an insufficient understanding of medical protocols when compared to human experts. The study also found a strong correlation between the length of output and the error rate of LLMs; that is, results from LLMs that were longer were more likely to contain incorrect diagnoses or medical details.

2.3.2 LLM-detection for Swedish text

A recent study investigated LLM reasoning in Swedish. Moell and Boye (2025) studied the performance of LLMs on language complexity measurement tasks (LIX readability metric and Average Dependency Distance). The results of the study found that all models demonstrated at least some capacity for each of these tasks, but that LLMs remain prone to errors in explicit calculations and syntactic analysis.

- discuss differences in text stylistic

2.3.3 The LLMs of focus for this thesis

1. OpenAI GPT-5
2. Google Gemini 3 Flash
3. GPT-SW3

2.4 Background summary

3 Methodology

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4 Results

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5 Discussion

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6 Conclusion

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